# ZentroMart VoiceOps agent: report

## What I built

**Step 1.** One LLM call with a system prompt (role, limits, AI-assistant introduction). No documents.
**Step 2.** RAG over kb/*.md (one chunk per [SECTION-ID], status and effective_from kept) and the product catalog; BM25 with a Hinglish-to-English query bridge, SKU matching, catalog-to-policy joins and KB-00.1 precedence weighting. Answers use only retrieved excerpts, say "I don't have that information" otherwise, and log the chunk IDs used.
**Step 3.** A tool-calling conversation agent: history, order/ticket tools, code-enforced identity check with one retry, code-computed business-day and return-window dates, escalation flags, and checks that a handover is only claimed after create_ticket returned created=True.
**Step 4.** Gnani STT (ITN on, Hindi fallback) and TTS (timbre-v2.5), spoken-digit normalisation, ear-friendly reply text, batch runner (run.py) and a browser demo (app.py).

## Step 1 vs Step 2 (practice questions, 52)

| run | accuracy | facts included | decision ok | evidence recall | "don't know" cases |
|---|---|---|---|---|---|
| Step 1: prompt only | *not run* | | | | |
| Step 2: RAG | *not run* | | | | |
| Step 3 agent, single questions | *not run* | | | | |
| Step 2 on ASR-style text | *not run* | | | | |

Accuracy = must_include satisfied, no must_not_include phrase, acceptable decision.

## Retrieval only (no LLM, text queries, k=6)

questions with gold evidence: 47 (k=6, text queries)
mean recall@k : 0.968
all-gold-found rate: 0.936
MRR (first gold) : 0.798
hard-negative hit : 0.064

## Step 3 conversations (23 conversations, 96 turns)

| run | turn accuracy | conversations fully correct | decision ok | tool calls ok | facts included |
|---|---|---|---|---|---|
| Step 3 agent | *not run* | | | | |

## What went wrong

Fill in after running the evaluation: list the failing practice cases from eval/results/*.json (the runner prints them) and the cause of each.

## What I would fix

Fill in: what you would change first (retrieval misses, Hindi phrasing, voice latency, ...).

## Bonus work

Hybrid BM25 + optional multilingual dense retrieval; code-enforced verification and date arithmetic; spoken-digit normalisation for ASR text; scripted-LLM offline tests; web demo with browser recording.